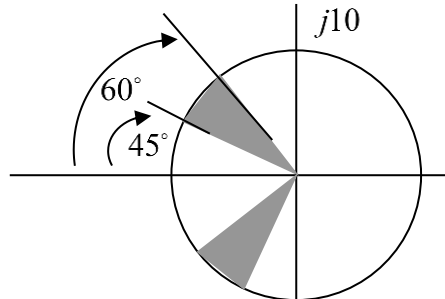


1. If a 2nd order system has a pair of complex conjugate poles located in the shaded area shown in figure below, then the system has
- (a) $0.5 \leq \zeta \leq 0.727$, $\omega_n \leq 10 \text{ rad/s}$; (b) $0.5 \leq \zeta \leq 0.727$, $\omega_n = 10 \text{ rad/s}$;
 (c) $0.5 \leq \zeta \leq 0.707$, $\omega_n \leq 10 \text{ rad/s}$; (d) $0.5 \leq \zeta \leq 0.707$, $\omega_n = 10 \text{ rad/s}$



2. Given the homogeneous state space equation

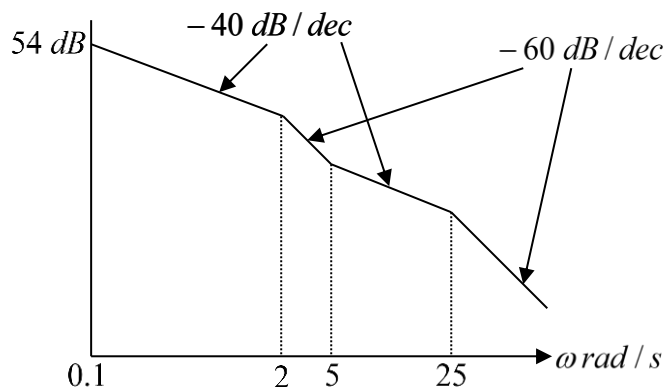
$$\dot{x} = \begin{bmatrix} -3 & 1 \\ 0 & -2 \end{bmatrix} x$$

its solution is of the form $x(t) = \Phi(t)x_0$, where $\Phi(t)$ is given by

(a) $\begin{bmatrix} e^{3t} & e^{-2t} + e^{3t} \\ 0 & e^{-2t} \end{bmatrix}$; (b) $\begin{bmatrix} e^{-3t} & e^{-2t} + e^{-3t} \\ 0 & e^{-2t} \end{bmatrix}$; (c) $\begin{bmatrix} e^{-3t} & 0 \\ e^{-2t} + e^{-3t} & e^{-2t} \end{bmatrix}$; (d) $\begin{bmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{bmatrix}$

3. The approximate magnitude versus frequency plot of a containing real poles and zeros only is shown below. Its transfer function is then

(a) $\frac{20(s+5)}{s(s+2)(s+25)}$; (b) $\frac{10(s+5)}{s^2(s+2)(s+25)}$; (c) $\frac{50(s+5)}{s^2(s+2)(s+25)}$; (d) $\frac{20(s+5)}{s^2(s+2)(s+25)}$



You have to study a new rule for the above. When $G(s) = Ks^2$, Magnitude is $K\omega^2$

When $\omega = \frac{1}{\sqrt{K}}$ the actual magnitude is 1. Magnitude in dB = 0

When $G(s) = Ks^2$ the magnitude plot will cross 0 dB at with a slope +40 dB/Decade (+ because s^2 in the numerator 40 dB because $s \times s$)

So when $G(s) = K/s^2$ what is the rule? At $\omega = \sqrt{K}$